

# MetaOB Trend-Filter Validation

No Filter vs. the Live (Buggy) Sharpe Filter vs. the D1-Regression Fix, by Asset Class

46 instruments across 8 asset-class buckets | M15 entries, D1 trend | Generated 2026-07-23 18:58 UTC

## Executive Summary

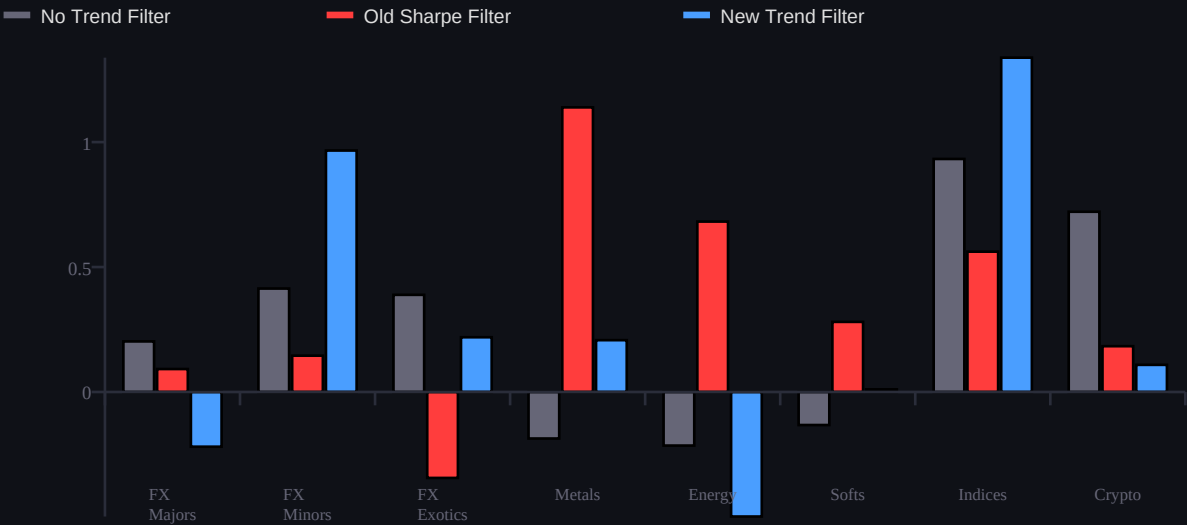
A live-trade audit found MetaOB entries firing against the genuine long-term (D1) trend: the **Old Sharpe Filter** that was actually running in production classified trend using a rolling Sharpe ratio of M15 returns over a window that, despite being named *sma\_long\_hours*, was only ~190-220 M15 bars (~2 days), thresholded against a quantile recalibrated daily from the trailing 66 days. The **New Trend Filter** fix replaces this with a rolling OLS t-stat of D1 closes over a 200-bar (~200-trading-day) window - a genuine long-term trend read. **No Trend Filter** (the raw OB + pivot-breakout signal, no trend gate at all) is included as the baseline both other configs are gating.

**Headline finding:** **New Trend Filter** has the highest average Sharpe across the full universe (+0.38). Head-to-head against the filter that was actually live, the New Trend Filter beats Old Sharpe Filter on **27/46 symbols** (19 favor the old filter, 0 roughly tied), for an average Sharpe delta of +0.04. The fix corrects a real trend-alignment bug (confirmed independently against D1 SMA200 on live trade history) but that is not automatically the same claim as "higher Sharpe everywhere" - see the per-symbol head-to-head in Section 3.

| Config            | Sharpe (vbt) | Sharpe (qs) | Avg Edge | Win Rate | Avg PF | Trades/Sym | Sharpe+ % |
|-------------------|--------------|-------------|----------|----------|--------|------------|-----------|
| No Trend Filter   | +0.32        | +0.22       | +0.4pp   | 33.8%    | 1.03   | 647.5      | 67%       |
| Old Sharpe Filter | +0.33        | +0.21       | +2.6pp   | 35.9%    | 1.24   | 73.2       | 65%       |
| New Trend Filter  | +0.38        | +0.26       | +2.0pp   | 35.3%    | 1.11   | 275.5      | 67%       |

## 2. Performance by Asset-Class Bucket

Average Sharpe Ratio by Bucket and Configuration



| Bucket     | Winner            | Winner Sharpe | No-Filter Sharpe | N Symbols | Closed Trades (all cfigs) |
|------------|-------------------|---------------|------------------|-----------|---------------------------|
| FX Majors  | No Trend Filter   | +0.20         | +0.20            | 7         | 8148                      |
| FX Minors  | New Trend Filter  | +0.97         | +0.41            | 9         | 9879                      |
| FX Exotics | No Trend Filter   | +0.39         | +0.39            | 3         | 3094                      |
| Metals     | Old Sharpe Filter | +1.14         | -0.19            | 5         | 4999                      |
| Energy     | Old Sharpe Filter | +0.68         | -0.21            | 3         | 2930                      |
| Softs      | Old Sharpe Filter | +0.28         | -0.13            | 6         | 3228                      |
| Indices    | New Trend Filter  | +1.34         | +0.93            | 7         | 7657                      |
| Crypto     | No Trend Filter   | +0.72         | +0.72            | 6         | 5892                      |

### 3. Per-Symbol Head-to-Head: New Trend Filter vs. Old Sharpe Filter

Sorted by Sharpe delta (New - Old). 27 symbols favor the fix, 19 favor the filter that was actually live, 0 are roughly tied ( $|\Delta| \leq 0.05$ ).

| Symbol | Bucket     | Old Sharpe | New Sharpe | $\Delta$ (New-Old) |
|--------|------------|------------|------------|--------------------|
| NZDCHF | FX Minors  | -0.60      | +2.30      | +2.90              |
| GBPJPY | FX Minors  | -0.22      | +1.59      | +1.82              |
| UK100  | Indices    | +0.04      | +1.71      | +1.67              |
| CADCHF | FX Minors  | +0.16      | +1.69      | +1.52              |
| EURNOK | FX Exotics | -1.04      | +0.35      | +1.39              |
| SUGAR  | Softs      | -0.09      | +1.04      | +1.13              |
| GER40  | Indices    | +0.90      | +1.85      | +0.95              |
| USDZAR | FX Exotics | -0.50      | +0.43      | +0.93              |
| BTCUSD | Crypto     | -0.47      | +0.38      | +0.84              |
| EURJPY | FX Minors  | +0.60      | +1.38      | +0.79              |
| EURCHF | FX Minors  | -0.06      | +0.71      | +0.78              |
| JP225  | Indices    | +0.55      | +1.32      | +0.77              |
| US100  | Indices    | +0.22      | +0.98      | +0.76              |
| XAUUSD | Metals     | +0.42      | +1.13      | +0.72              |
| AUDUSD | FX Majors  | -0.83      | -0.21      | +0.62              |
| USDJPY | FX Majors  | +0.37      | +0.97      | +0.59              |
| US30   | Indices    | -0.06      | +0.46      | +0.52              |
| HK50   | Indices    | +1.65      | +2.15      | +0.50              |
| GBPCHF | FX Minors  | +0.41      | +0.90      | +0.48              |
| XNGUSD | Energy     | -0.21      | +0.21      | +0.41              |
| XCUUSD | Metals     | +1.30      | +1.70      | +0.39              |
| COTTON | Softs      | +0.01      | +0.39      | +0.38              |
| WHEAT  | Softs      | -0.10      | +0.20      | +0.31              |
| US500  | Indices    | +0.63      | +0.90      | +0.27              |
| NZDUSD | FX Majors  | +0.14      | +0.39      | +0.25              |
| ADAUSD | Crypto     | +0.58      | +0.81      | +0.23              |
| EURGBP | FX Minors  | -0.35      | -0.14      | +0.22              |
| ETHUSD | Crypto     | -0.26      | -0.47      | -0.21              |
| DOGUSD | Crypto     | +0.60      | +0.38      | -0.22              |

| Symbol | Bucket     | Old Sharpe | New Sharpe | Δ (New-Old) |
|--------|------------|------------|------------|-------------|
| USDSGD | FX Minors  | +0.07      | -0.18      | -0.25       |
| USDCHE | FX Majors  | +0.45      | +0.07      | -0.38       |
| LTCUSD | Crypto     | -0.40      | -0.89      | -0.49       |
| SOLUSD | Crypto     | +1.04      | +0.45      | -0.59       |
| USDCNH | FX Exotics | +0.50      | -0.12      | -0.63       |
| CORN   | Softs      | -0.49      | -1.15      | -0.65       |
| GBPUSD | FX Majors  | -1.32      | -2.01      | -0.69       |
| COCOA  | Softs      | +2.05      | +1.29      | -0.77       |
| XBRUSD | Energy     | +0.13      | -0.69      | -0.81       |
| AUDJPY | FX Minors  | +1.30      | +0.44      | -0.86       |
| USDCAD | FX Majors  | +0.93      | -0.03      | -0.95       |
| XPTUSD | Metals     | +0.15      | -0.97      | -1.12       |
| XAGUSD | Metals     | +2.13      | +0.80      | -1.33       |
| EURUSD | FX Majors  | +0.90      | -0.71      | -1.61       |
| COFFEE | Softs      | +0.31      | -1.71      | -2.02       |
| WTI    | Energy     | +2.12      | -1.02      | -3.14       |
| XPDUSD | Metals     | +1.69      | -1.62      | -3.31       |

## 4. Methodology & Caveats

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- **Entry/exit model.** MetaOB places market orders with no concurrency cap live (the `are_positions_with_tag_open` guard in `signals()` is commented out) - this backtest mirrors that: a position opens on every bar the OB+pivot+trend condition holds, sized at a fixed 2% of current equity per trade (percent sizing keeps stats comparable across instruments at very different price scales - see `build_vbt_portfolio` in `metafvg_backtest.py`, reused unchanged here). No spread, slippage, or commission is modeled.
- **Same OB/pivot/ATR parameters across the whole universe** (`pivot_window=60`, `breakout_lookback=3`, `atr_period=14`, `sl_atr_mult=5`, `tp_atr_mult=10`) - not per-symbol-tuned like the live prod config (`metaob.yaml`). This isolates the trend-filter effect from per-symbol parameter tuning, but means absolute Sharpe/PF here should not be read as "what prod actually earns" for any one symbol.
- **New Trend Filter uses only fully-closed D1 bars** - a day's regression output becomes available starting the following calendar day. Live `MetaOB.compute_trend_t()` can also see the still-forming current day's D1 bar, so this backtest is slightly more conservative (up to one day slower to react) than live.
- **Old Sharpe Filter** replicates the original production logic faithfully, including its daily threshold recalibration (mirrors the old `fit()`, rescheduled daily at 00:01 UTC by `metaworker.py`) from a trailing 66-day quantile of the rolling-Sharpe series - not a single fixed threshold for the whole backtest.
- **2-year window, M15 entries.** Both configs share the same fetched OHLC per symbol, so differences are attributable to the trend filter alone, not to different market windows.
- **quantstats metrics** (Sharpe, Sortino, Calmar, CAGR, max drawdown) are computed on the `vectorbt` equity curve resampled to daily, per `quantstats`' own annualization convention - reported per (symbol, config) in the underlying cache (`metaob_ab_sweep_data.pkl`) even though only vbt Sharpe and qs Sharpe are tabulated here for space.
- **Sample size.** Per-symbol trade counts vary widely by config (the trend filters are, by design, selective) - see the "Trades/Sym" column in Section 1. Individual symbol Sharpe figures with few closed trades should be read with proportionally more skepticism.